

KAGGLE · BUILD WITH AI — GEMMA HACKATHON · GDG WINDSOR · EDGE / ON-DEVICE TRACK

GEMMA MONSTERS

A monster-battling castle with a real tutoring agent inside.

Five monsters guard the Citadel. Each one is a trick — a wrong idea that feels right.

Powered by Gemma, running entirely on one laptop. Nothing a student does ever leaves it.

Gokulakrishnan · Padmanabha · Taha · Naimah · Amarah

Wrong answers are not random

$$\frac{2}{3} + \frac{1}{4} = \frac{3}{7}$$

Tops with tops, bottoms with bottoms. Feels tidy. Completely wrong.

That is a trick: the wrong idea that feels right.

This student does not need more practice — they need that one idea found and removed. Practice apps mark the answer wrong and move on. Cloud tutors need the internet, ship a child's work to a server, and sometimes invent mathematics.

So we turned every trick into a monster

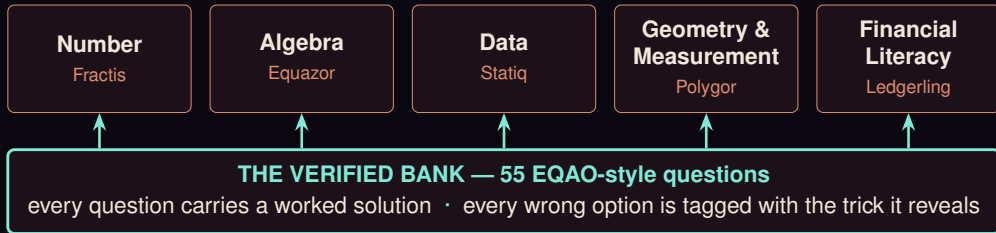
A gothic castle built in three.js. Five monsters on floating platforms, each one *being* a trick.

Monster	Guards the strand	The trick it plants in your head
Fractis	Number	“just add fractions straight across”
Equazor	Algebra	signs that flip when crossing the equals sign
Statiq	Data	mean and median blurred into one fuzzy word
Polygor	Geometry & Measurement	stolen formulas that <i>almost</i> fit
Ledgerling	Financial Literacy	interest quietly skimmed off a sloppy budget

You beat a monster by proving its trick no longer fools you — never by luck.

Beat all five and the golden gate opens: someone is locked in the keep — the rescue finale.

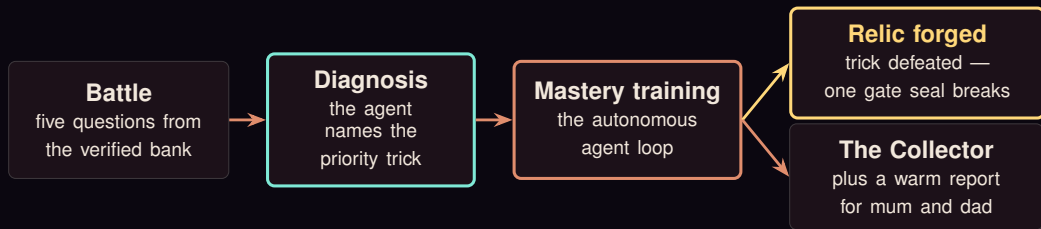
Built on the real Grade 9 curriculum



So when a student picks a wrong answer, **the wrong answer itself is the diagnosis** — a table lookup, not a model guess.

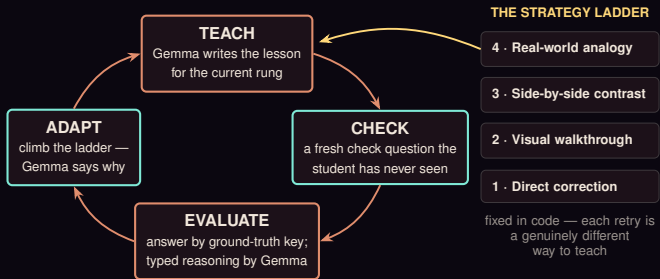
Every item maps to a published MTH1W expectation and **sits inside what the EQAO assessment covers** — in Financial Literacy, students **compare** the effects of rates and payment terms.

One session, start to finish



- The monster meets you **by name** — and it **remembers**: Gemma writes its greeting from your real battle history.
- Between battles: **90-second mental-math skirmishes** against the Collector's lieutenants, and his three-life **boss speed trial**.
- Defeat all five monsters → the golden gate opens → **the rescue finale**.

The agent loop: teach until the trick is gone



Teal = code decides. Coral = Gemma judges.

Deterministic spine: plain code runs the loop; Gemma is called only at the judgment points.

Two exits, no third: two fresh correct answers with solid reasoning **forges a relic**; a spent ladder means a **warm report for mum and dad**. The agent never gives up silently.

Every switch of strategy is announced on screen, with its reason.

A right answer is not enough

After every check question, the student must **type why** — and Gemma grades those words.

THE CHECK QUESTION

Calculate $\frac{1}{2} + \frac{1}{3}$ — picks $\frac{5}{6}$ (correct)

THE STUDENT TYPES

"I just guessed the biggest fraction"

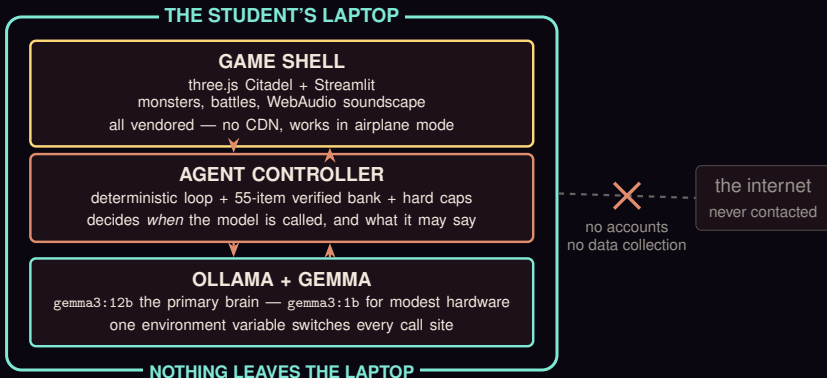
GEMMA'S GRADE

SHALLOW — does **not** count toward mastery

- Three labels only — **RESOLVED** / **SHALLOW** / **SAME_ERROR**. A closed set, nothing open-ended.
- The monster **answers the student's own words back, in character** — it read what you wrote.
- **Fail-open**: if the grader hiccups, the student is never punished for it.

A lucky bubble cannot fake understanding past this gate.

Everything on one laptop



The Edge / On-Device promise, kept literally: **a student's struggles stay on their own machine.**

Gemma judges. Code guarantees.

WHAT GEMMA DECIDES

the judgment calls

writes every lesson and study guide
grades your typed reasoning
picks the next rung — and says why
authors fresh check questions (audited)
voices the monsters; forges relics
phrases the report for mum and dad

WHAT CODE GUARANTEES

the ground truth

all math facts from the verified bank
answers graded by ground-truth key
the trick diagnosed by tag lookup
the ladder order is fixed
hard caps end every session
scores computed in code, injected

THE GROUNDING RULE — the verified worked solution rides inside every prompt:

Gemma explains *why*; it never recomputes the math and never states new numbers.

The model does the thinking — **it is never the source of mathematical truth.**

We planned for the model to fail

Every guardrail exists because live testing broke something first.

Fresh check questions, self-audited

The small model once wrote a question with a wrong answer key. Now: bank first; a generated check question ships only if Gemma can **solve it blind** and agree with its own key.

No invented arithmetic

The 1B once produced wrong math mid-explanation. Now the model never computes at all: the verified solution is the **only math it may state**.

Hard caps — it always ends

4 check cycles, a 12-model-call budget, and ladder exhaustion each force a clean hand-off.
The loop always terminates.

The warm hand-off

When teaching stalls, mum and dad get a **warm report**: the trick, what was tried, and **kitchen-table activities** to do together — a hand-off to a human, never a silent stop.

The guardrails carry correctness; model size only dials quality.

What we would want a judge to notice

GEMMA IS THE ENGINE

Ten call sites behind one auditable door: lessons, reasoning grades, strategy picks, check questions, coaching, relics, reports, and the director that decides where the student goes next.

Fully local through Ollama — `gemma3:12b`, with 1b one flag away.

NOBODY HAS BUILT THIS

A tutoring agent you can *fight*: every trick embodied as a monster that has to be beaten on understanding.

Mastery is gated on the student's own typed reasoning — and the citadel answers those words back.

IT ACTUALLY RUNS

Live and end to end with the Wi-Fi off: citadel, battles, the agent loop, speed skirmishes, the boss trial, the rescue.

Hard caps mean no session can hang, and a model hiccup never costs the student anything.

WE SHOW OUR WORK

An honest engineering story: what the model broke under testing, and the structural fix behind each guardrail.

Every item in the bank maps to a published Grade 9 expectation.

Every monster is a trick — a wrong idea that feels right.

Defeat the idea, and the gate opens.

A student walks in adding fractions straight across. Twenty minutes later they have *proven* the trick is gone — and nothing about them ever left their laptop.

github.com/EdTechDL/gemma-without-borders · live demo, Wi-Fi off · companion
Kaggle notebook

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